

Chatbot Development Using Hugging Face Transformers

- **An AI-Powered Document Analyzer using Retrieval-Augmented Generation (RAG)**

Project Proposal Submitted

To

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Sr. No	Index	Page no
1.	Introduction	3
2.	Objectives of the Study	3
3.	Scope of the Project	4
4.	Work Plan and Methodology	5
5.	References	6

1. Introduction

The proposed project, titled "Chatbot Development Using Hugging Face Transformers - An AI-Powered Document Analyzer using Retrieval-Augmented Generation (RAG)", is related to the AI & ML specialization. It focuses on developing an intelligent chatbot that can read and understand uploaded PDF documents and answer user questions based on the content of those documents.

In many academic and business environments, users need to search through long reports, manuals, notes, research papers, policies, and other PDF documents. Manual searching is time-consuming and often inefficient, especially when the user does not know the exact keywords used in the document. A normal keyword search may only match exact words, while the user may ask the same question in a different way.

This project addresses that problem by applying Natural Language Processing, semantic search, vector databases, and Hugging Face Transformer models. The system will use a Hugging Face sentence-transformer model to convert document text into semantic embeddings. These embeddings represent the meaning of the text in numerical form. FAISS will be used to store and search these embeddings efficiently. When a user asks a question, the system retrieves the most relevant sections of the uploaded document and uses a Hugging Face Transformer-based question-answering or text-generation model to generate a context-aware answer.

The project follows the Retrieval-Augmented Generation approach. In this approach, the answer is not generated blindly. First, the system retrieves relevant information from the document, and then the LLM generates the final answer using that retrieved context. This makes the chatbot more useful for document-based question answering and reduces the chances of irrelevant or unsupported responses.

The full-stack application will include a React frontend for document upload and chat interaction, and a FastAPI backend for PDF processing, embedding generation, vector search, and Hugging Face model inference. The project is practical, industry-relevant, and suitable for demonstrating the application of AI and ML concepts in a real-world document analysis use case.

2. Objectives of the Study

The main objective of this project is to design and develop an AI-powered document chatbot that can help users interact with PDF documents through natural language questions. The key objectives are as follows:

- To develop an AI-powered chatbot using Hugging Face Transformers for intelligent document-based question answering.
- To implement Retrieval-Augmented Generation (RAG) so that the chatbot can retrieve relevant document content before generating answers.
- To enable users to upload PDF documents and ask questions in simple natural language.
- To use Hugging Face embedding models for converting document chunks and user questions into semantic vectors.
- To use FAISS for fast and efficient vector similarity search across document content.
- To integrate Hugging Face Transformer-based models for document-grounded answer generation without depending on paid external APIs.

- To build a simple and user-friendly full-stack application using React and FastAPI.
- To reduce manual document searching by providing fast, context-aware, and intelligent answers from uploaded files.

These objectives are designed to make the project academically relevant as well as practically useful. The project demonstrates how AI and ML can improve document understanding, information retrieval, and user interaction with digital content.

3. Scope of the Project

The scope of the project includes the design and development of a document-based chatbot system for text-based PDF analysis. The system will allow a user to upload a PDF, process the document, ask questions, and receive answers based on the retrieved content.

Functional Scope

- PDF document upload through the web interface.
- Text extraction from uploaded PDF files.
- Dividing extracted text into smaller chunks for better processing.
- Generating semantic embeddings using Hugging Face Transformer models.
- Storing and searching document embeddings using FAISS.
- Retrieving relevant document chunks based on user questions.
- Generating context-aware answers using a Hugging Face Transformer-based model.
- Displaying answers in a chatbot-style interface.

Technical Scope

- Frontend development using React and Vite.
- Backend API development using FastAPI.
- AI and NLP workflow using Hugging Face Transformers, FAISS, and RAG.
- Local model execution to reduce dependency on paid external APIs.
- Basic testing of response relevance, retrieval quality, and application usability.

Limitations

- The system will mainly support text-based PDF documents.
- It will not focus on handwritten document recognition or image-based OCR in the initial version.
- Response quality may depend on the clarity of the uploaded document, chunking strategy, retrieval quality, and the selected Hugging Face model.
- Large documents may require more processing time depending on system hardware.

4. Work Plan and Methodology

The project will be developed using a phased methodology. Each phase will focus on a specific part of the system, from requirement analysis to testing and documentation.

System Workflow

- User uploads a PDF document through the React frontend.
- The FastAPI backend receives the file and extracts text from the PDF.
- The extracted text is cleaned and divided into smaller chunks.
- Hugging Face Transformer models generate semantic embeddings for each chunk.
- FAISS stores the embeddings and performs similarity search.
- When the user asks a question, the question is also converted into an embedding.
- FAISS retrieves the most relevant chunks from the uploaded document.
- The retrieved chunks and user question are passed to the Hugging Face Transformer-based answer generation model.
- The LLM generates a context-aware answer, which is displayed in the chatbot interface.

Methodology Phases

Phase 1: Requirement Analysis and Planning: Understand project requirements, define the problem statement, finalize the project objectives, and identify the required tools and technologies.

Phase 2: System Design: Prepare the overall system architecture, decide the document processing workflow, and plan frontend-backend communication.

Phase 3: Frontend Development: Create a simple React interface for PDF upload, chat input, and answer display. The interface will focus on usability and clarity.

Phase 4: Backend Development: Build FastAPI endpoints for file upload, text extraction, embedding generation, vector search, and answer generation.

Phase 5: PDF Processing and Chunking: Extract text from PDF files and divide it into smaller chunks so that the retrieval model can search document content accurately.

Phase 6: Embedding and Vector Search: Use Hugging Face Transformer models to generate embeddings and FAISS to store and retrieve semantically similar document chunks.

Phase 7: Hugging Face Model Integration: Connect the retrieved context with a Hugging Face Transformer-based model to generate final answers based on the uploaded document.

Phase 8: Testing and Documentation: Test the application using sample PDFs, verify answer relevance, fix errors, and prepare project documentation and presentation material.

Technology Stack

Component	Technology / Tool
Frontend	React with Vite
Backend	FastAPI
Programming Language	Python and JavaScript
Embedding Model	Hugging Face Transformer (sentence-transformers)
Vector Search	FAISS
LLM	Hugging Face Transformer (text-generation model)
Document Processing	PyPDF / PDFPlumber
Application Type	Full-stack AI document chatbot application

Expected Outcomes

- A working AI-powered PDF question-answering chatbot.
- A practical demonstration of Hugging Face Transformers in document understanding, semantic search, and answer generation.
- Implementation of RAG architecture using Hugging Face embeddings, FAISS vector search, and Transformer-based response generation.
- A user-friendly interface for uploading documents and asking questions.
- Reduced manual effort in searching and understanding large PDF documents.

5. References

- Vaswani, Ashish, et al. "Attention Is All You Need." Advances in Neural Information Processing Systems, 2017.
- Lewis, Patrick, et al. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." Advances in Neural Information Processing Systems, 2020.
- Reimers, Nils, and Iryna Gurevych. "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks." EMNLP-IJCNLP, 2019.
- Johnson, Jeff, Matthijs Douze, and Herve Jegou. "Billion-scale similarity search with GPUs." IEEE Transactions on Big Data, 2019.
- Hugging Face Transformers Documentation. <https://huggingface.co/docs/transformers>
- [FAISS Documentation](#)
- Hugging Face Model Hub and Pipelines Documentation.
- [FastAPI Documentation.](#)
- [React Documentation.](#)